```
!pip install transformers[sentencepiece] datasets sacrebleu rouge_score py7zr -q
```

```
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 7.9/7.9 MB 31.4 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 493.7/493.7 kB 38.2 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 119.7/119.7 kB 17.4 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.0/67.0 kB 9.2 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 311.1/311.1 kB 34.2 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.8/3.8 MB 75.7 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.3/1.3 MB 62.5 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.3/1.3 MB 70.3 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 115.3/115.3 kB 16.4 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.8/134.8 kB 15.5 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.1/2.1 MB 81.0 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 412.3/412.3 kB 43.4 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 138.9/138.9 kB 17.2 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 49.7/49.7 kB 7.4 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 93.1/93.1 kB 14.1 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.0/3.0 MB 85.8 MB/s eta 0:00:00
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 295.0/295.0 kB 36.0 MB/s eta 0:00:00
  Building wheel for rouge_score (setup.py) ... done
```

```
!nvidia-smi
```

```
Tue Nov 14 09:39:16 2023
+-----------------------------------------------------------------------------+
| NVIDIA-SMI 525.105.17   Driver Version: 525.105.17   CUDA Version: 12.0     |
|-------------------------------+----------------------+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|                               |                      |               MIG M. |
|===============================+======================+======================|
|   0  Tesla T4            Off  | 00000000:00:04.0 Off |                    0 |
| N/A   55C    P8     9W /  70W |      0MiB / 15360MiB |      0%      Default |
|                               |                      |                  N/A |
+-------------------------------+----------------------+----------------------+

+-----------------------------------------------------------------------------+
| Processes:                                                                  |
|  GPU   GI   CI        PID   Type   Process name                  GPU Memory |
|        ID   ID                                                   Usage      |
|=============================================================================|
|  No running processes found                                                 |
+-----------------------------------------------------------------------------+
```

```
!pip install -U accelerate==0.20.3
```

```
Collecting accelerate==0.20.3
  Downloading accelerate-0.20.3-py3-none-any.whl (227 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 227.6/227.6 kB 5.3 MB/s eta 0:00:00
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.10/dist-packages (from accelerate==0.20.3) (1.23.5)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.10/dist-packages (from accelerate==0.20.3) (23.2)
Requirement already satisfied: psutil in /usr/local/lib/python3.10/dist-packages (from accelerate==0.20.3) (5.9.5)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.10/dist-packages (from accelerate==0.20.3) (6.0.1)
Requirement already satisfied: torch>=1.6.0 in /usr/local/lib/python3.10/dist-packages (from accelerate==0.20.3) (2.1.0+cu118)
Requirement already satisfied: filelock in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (3.13.1)
Requirement already satisfied: typing-extensions in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (4.
Requirement already satisfied: sympy in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (1.12)
Requirement already satisfied: networkx in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (3.2.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (3.1.2)
Requirement already satisfied: fsspec in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (2023.6.0)
Requirement already satisfied: triton==2.1.0 in /usr/local/lib/python3.10/dist-packages (from torch>=1.6.0->accelerate==0.20.3) (2.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.10/dist-packages (from jinja2->torch>=1.6.0->accelerate==0.20.
Requirement already satisfied: mpmath>=0.19 in /usr/local/lib/python3.10/dist-packages (from sympy->torch>=1.6.0->accelerate==0.20.3) (
Installing collected packages: accelerate
Successfully installed accelerate-0.20.3
```

```
from transformers import pipeline, set_seed

import matplotlib.pyplot as plt
from datasets import load_dataset
import pandas as pd
from datasets import load_dataset, load_metric

from transformers import AutoModelForSeq2SeqLM, AutoTokenizer
```

```
import nltk
from nltk.tokenize import sent_tokenize

from tqdm import tqdm
import torch

nltk.download("punkt")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
True
```

```
from transformers import AutoModelForSeq2SeqLM, AutoTokenizer

device = "cuda" if torch.cuda.is_available() else "cpu"

model_ckpt = "google/pegasus-cnn_dailymail"

tokenizer = AutoTokenizer.from_pretrained(model_ckpt)

model_pegasus = AutoModelForSeq2SeqLM.from_pretrained(model_ckpt).to(device)
```

```
Downloading (…)okenizer_config.json: 100%  88.0/88.0 [00:00<00:00, 6.30kB/s]
Downloading (…)lve/main/config.json: 100%  1.12k/1.12k [00:00<00:00, 71.0kB/s]
Downloading (…)ve/main/spiece.model: 100%  1.91M/1.91M [00:00<00:00, 4.65MB/s]
Downloading (…)cial_tokens_map.json: 100%  65.0/65.0 [00:00<00:00, 3.06kB/s]
Downloading pytorch_model.bin: 100%  2.28G/2.28G [00:12<00:00, 175MB/s]
Some weights of PegasusForConditionalGeneration were not initialized from the model checkpoint at google/pegasus-cnn_dailymail and are
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Downloading (…)neration_config.json: 100%  280/280 [00:00<00:00, 4.71kB/s]
```

```
def generate_batch_sized_chunks(list_of_elements, batch_size):
    """split the dataset into smaller batches that we can process simultaneously
    Yield successive batch-sized chunks from list_of_elements."""
    for i in range(0, len(list_of_elements), batch_size):
        yield list_of_elements[i : i + batch_size]


def calculate_metric_on_test_ds(dataset, metric, model, tokenizer,
                                batch_size=16, device=device,
                                column_text="article",
                                column_summary="highlights"):
    article_batches = list(generate_batch_sized_chunks(dataset[column_text], batch_size))
    target_batches = list(generate_batch_sized_chunks(dataset[column_summary], batch_size))

    for article_batch, target_batch in tqdm(
        zip(article_batches, target_batches), total=len(article_batches)):

        inputs = tokenizer(article_batch, max_length=1024,  truncation=True,
                        padding="max_length", return_tensors="pt")

        summaries = model.generate(input_ids=inputs["input_ids"].to(device),
                         attention_mask=inputs["attention_mask"].to(device),
                         length_penalty=0.8, num_beams=8, max_length=128)
        ''' parameter for length penalty ensures that the model does not generate sequences that are too long. '''

        # Finally, we decode the generated texts,
        # replace the  token, and add the decoded texts with the references to the metric.
        decoded_summaries = [tokenizer.decode(s, skip_special_tokens=True,
                                clean_up_tokenization_spaces=True)
               for s in summaries]

        decoded_summaries = [d.replace("", " ") for d in decoded_summaries]


        metric.add_batch(predictions=decoded_summaries, references=target_batch)

    #  Finally compute and return the ROUGE scores.
    score = metric.compute()
    return score
```

Load data

Link: https://huggingface.co/datasets/samsum

```
dataset_samsum = load_dataset("samsum")

split_lengths = [len(dataset_samsum[split])for split in dataset_samsum]

print(f"Split lengths: {split_lengths}")
print(f"Features: {dataset_samsum['train'].column_names}")
print("\nDialogue:")

print(dataset_samsum["test"][1]["dialogue"])

print("\nSummary:")

print(dataset_samsum["test"][1]["summary"])
```

```
Downloading builder script: 100%  3.36k/3.36k [00:00<00:00, 140kB/s]
Downloading metadata: 100%  1.58k/1.58k [00:00<00:00, 99.8kB/s]
Downloading readme: 100%  7.04k/7.04k [00:00<00:00, 396kB/s]
Downloading data: 100%  2.94M/2.94M [00:00<00:00, 24.0MB/s]
Generating train split: 100%  14732/14732 [00:00<00:00, 22120.31 examples/s]
Generating test split: 100%  819/819 [00:00<00:00, 3.62 examples/s]
Generating validation split: 100%  818/818 [00:00<00:00, 3.61 examples/s]
Split lengths: [14732, 819, 818]
Features: ['id', 'dialogue', 'summary']

Dialogue:
Eric: MACHINE!
Rob: That's so gr8!
Eric: I know! And shows how Americans see Russian ;)
Rob: And it's really funny!
Eric: I know! I especially like the train part!
Rob: Hahaha! No one talks to the machine like that!
Eric: Is this his only stand-up?
Rob: Idk. I'll check.
Eric: Sure.
Rob: Turns out no! There are some of his stand-ups on youtube.
Eric: Gr8! I'll watch them now!
Rob: Me too!
Eric: MACHINE!
Rob: MACHINE!
Eric: TTYL?
Rob: Sure :)

Summary:
Eric and Rob are going to watch a stand-up on youtube.
```

Evaluating PEGASUS on SAMSum

```
dataset_samsum['test'][0]['dialogue']
```

```
'Hannah: Hey, do you have Betty's number?\nAmanda: Lemme check\nHannah: <file_gif>\nAmanda: Sorry, can't find it.\nAmanda: Ask Larry\n
Amanda: He called her last time we were at the park together\nHannah: I don't know him well\nHannah: <file_gif>\nAmanda: Don't be shy,
he's very nice\nHannah: If you say so..\nHannah: I'd rather you texted him\nAmanda: Just text him 🙂\nHannah: Urgh.. Alright\nHannah:
```

```
pipe = pipeline('summarization', model = model_ckpt )

pipe_out = pipe(dataset_samsum['test'][0]['dialogue'] )

print(pipe_out)
```

```
Some weights of PegasusForConditionalGeneration were not initialized from the model checkpoint at google/pegasus-cnn_dailymail and are
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Your max_length is set to 128, but your input_length is only 122. Since this is a summarization task, where outputs shorter than the in
[{'summary_text': "Amanda: Ask Larry Amanda: He called her last time we were at the park together .<n>Hannah: I'd rather you texted him
```

```
print(pipe_out[0]['summary_text'].replace(" .", ".\n"))
```

```
Amanda: Ask Larry Amanda: He called her last time we were at the park together.
<n>Hannah: I'd rather you texted him.
<n>Amanda: Just text him.
```

```
rouge_metric = load_metric('rouge')

score = calculate_metric_on_test_ds(dataset_samsum['test'], rouge_metric, model_pegasus, tokenizer, column_text = 'dialogue', column_summar
```

```
<ipython-input-12-388971e09d3a>:1: FutureWarning: load_metric is deprecated and will be removed in the next major version of datasets.
  rouge_metric = load_metric('rouge')
Downloading builder script:                                    5.65k/? [00:00<00:00, 225kB/s]
100%|██████████| 103/103 [18:04<00:00, 10.53s/it]
```

```
rouge_names = ["rouge1", "rouge2", "rougeL", "rougeLsum"]
rouge_dict = dict((rn, score[rn].mid.fmeasure ) for rn in rouge_names )

pd.DataFrame(rouge_dict, index = ['pegasus'])
```

	rouge1	rouge2	rougeL	rougeLsum
pegasus	0.015568	0.000299	0.015471	0.015497

Histogram

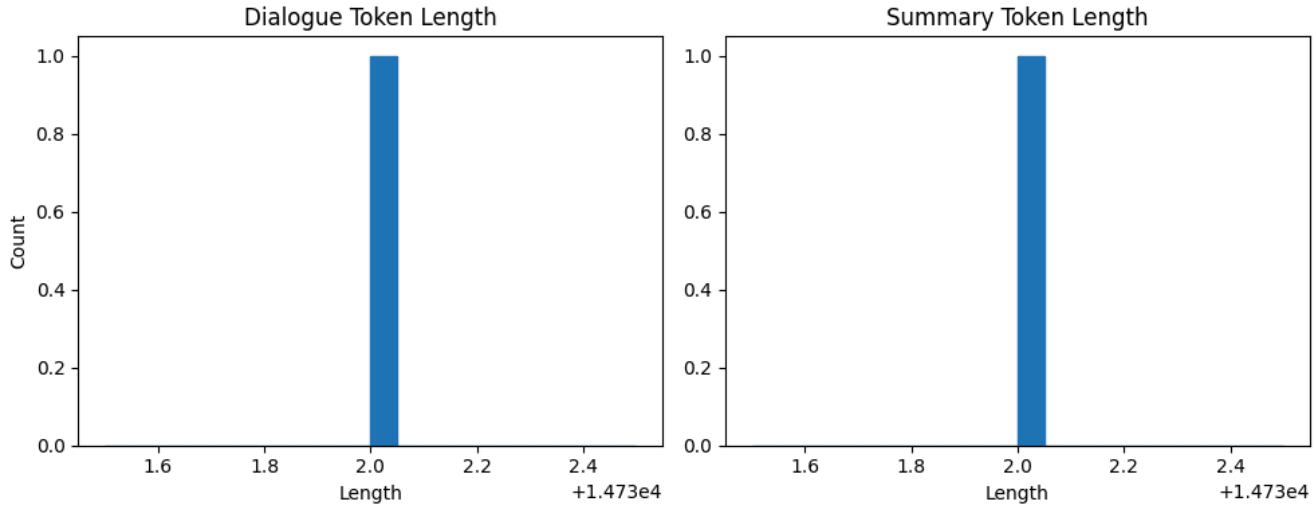
```
dialogue_token_len = len([tokenizer.encode(s) for s in dataset_samsum['train']['dialogue']])

summary_token_len = len([tokenizer.encode(s) for s in dataset_samsum['train']['summary']])
```

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
axes[0].hist(dialogue_token_len, bins = 20, color = 'C0', edgecolor = 'C0' )
axes[0].set_title("Dialogue Token Length")
axes[0].set_xlabel("Length")
axes[0].set_ylabel("Count")

axes[1].hist(summary_token_len, bins = 20, color = 'C0', edgecolor = 'C0' )
axes[1].set_title("Summary Token Length")
axes[1].set_xlabel("Length")
plt.tight_layout()
plt.show()
```

```
Token indices sequence length is longer than the specified maximum sequence length for this model (1044 > 1024). Running this sequence
```




```
def convert_examples_to_features(example_batch):
    input_encodings = tokenizer(example_batch['dialogue'] , max_length = 1024, truncation = True )
```

```
    with tokenizer.as_target_tokenizer():
        target_encodings = tokenizer(example_batch['summary'], max_length = 128, truncation = True )

    return {
        'input_ids' : input_encodings['input_ids'],
        'attention_mask': input_encodings['attention_mask'],
        'labels': target_encodings['input_ids']
    }

dataset_samsum_pt = dataset_samsum.map(convert_examples_to_features, batched = True)
```

```
Map: 100%                                                  14732/14732 [00:06<00:00, 2181.64 examples/s]
/usr/local/lib/python3.10/dist-packages/transformers/tokenization_utils_base.py:3856: UserWarning: `as_target_tokenizer` is deprecated
  warnings.warn(
Map: 100%                                                  819/819 [00:00<00:00, 2093.06 examples/s]
Map: 100%                                                  818/818 [00:00<00:00, 2116.83 examples/s]
```

```
from transformers import DataCollatorForSeq2Seq

seq2seq_data_collator = DataCollatorForSeq2Seq(tokenizer, model=model_pegasus)


from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
from transformers import TrainingArguments, Trainer

trainer_args = TrainingArguments(
    output_dir='pegasus-samsum', num_train_epochs=1, warmup_steps=500,
    per_device_train_batch_size=1, per_device_eval_batch_size=1,
    weight_decay=0.01, logging_steps=10,
    evaluation_strategy='steps', eval_steps=500, save_steps=1e6,
    gradient_accumulation_steps=16
)


trainer = Trainer(model=model_pegasus, args=trainer_args,
                  tokenizer=tokenizer, data_collator=seq2seq_data_collator,
                  train_dataset=dataset_samsum_pt["train"],
                  eval_dataset=dataset_samsum_pt["validation"])


trainer.train()
```

```
You're using a PegasusTokenizerFast tokenizer. Please note that with a fast tokenizer, using the `__call__` method is faster than using
[920/920 48:34, Epoch 0/1]
```

Step	Training Loss	Validation Loss
500	1.626500	1.484255

```
TrainOutput(global_step=920, training_loss=1.8238315105438232, metrics={'train_runtime': 2917.1108, 'train_samples_per_second': 5.05,
'train_steps_per_second': 0.315, 'total_flos': 5526698901602304.0, 'train_loss': 1.8238315105438232, 'epoch': 1.0})
```

```
score = calculate_metric_on_test_ds(
    dataset_samsum['test'], rouge_metric, trainer.model, tokenizer, batch_size = 2, column_text = 'dialogue', column_summary= 'summary'
)

rouge_dict = dict((rn, score[rn].mid.fmeasure ) for rn in rouge_names )

pd.DataFrame(rouge_dict, index = [f'pegasus'] )
```

```
100%|██████████| 410/410 [14:33<00:00,  2.13s/it]
```

	rouge1	rouge2	rougeL	rougeLsum
pegasus	0.018593	0.000324	0.018408	0.018457

```
# Save model
model_pegasus.save_pretrained("pegasus-samsum-model")
```

```
# Save tokenizer
tokenizer.save_pretrained("tokenizer")
```

```
('tokenizer/tokenizer_config.json',
 'tokenizer/special_tokens_map.json',
 'tokenizer/spiece.model',
 'tokenizer/added_tokens.json',
 'tokenizer/tokenizer.json')
```

Test

```
dataset_samsum = load_dataset("samsum")


tokenizer = AutoTokenizer.from_pretrained("tokenizer")


sample_text = dataset_samsum["test"][0]["dialogue"]

reference = dataset_samsum["test"][0]["summary"]


gen_kwargs = {"length_penalty": 0.8, "num_beams":8, "max_length": 128}

pipe = pipeline("summarization", model="pegasus-samsum-model",tokenizer=tokenizer)


print("Dialogue:")
print(sample_text)


print("\nReference Summary:")
print(reference)


print("\nModel Summary:")
print(pipe(sample_text, **gen_kwargs)[0]["summary_text"])
```

```
Your max_length is set to 128, but your input_length is only 122. Since this is a summarization task, where outputs shorter than the in
Dialogue:
Hannah: Hey, do you have Betty's number?
Amanda: Lemme check
Hannah: <file_gif>
Amanda: Sorry, can't find it.
Amanda: Ask Larry
Amanda: He called her last time we were at the park together
Hannah: I don't know him well
Hannah: <file_gif>
Amanda: Don't be shy, he's very nice
Hannah: If you say so..
Hannah: I'd rather you texted him
Amanda: Just text him 😊
Hannah: Urgh.. Alright
Hannah: Bye
Amanda: Bye bye

Reference Summary:
Hannah needs Betty's number but Amanda doesn't have it. She needs to contact Larry.

Model Summary:
Hannah is looking for Betty's number. Amanda can't find it. Larry called Betty last time they were at the park together. Hannah would r
```